KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6371A | **6 Credits: 4** | **Program Elective Course** | **Source-grounded**

Networks

● This course is designed to get a thorough knowledge of internal and external

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6371A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6371A.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	6371A
Course title	Networks
Semester	6 Credits: 4
Course category	Program Elective Course
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● This course is designed to get a thorough knowledge of internal and external
- components of desktop computers.
- ● Provide adequate knowledge of the concept of data communication and computer
- ● Acquire knowledge of data sharing, transmission media and protocols.

Course outcomes

CO	Official outcome	Study level
CO1	Outline basic structure of a computer, motherboard 14 Understanding components in desktop computers and BIOS settings	See official syllabus
CO2	Describe power supply and storage devices of a computer 14 Understanding Describe the basic concepts of data communication and	See official syllabus
CO3	15 Understanding computer networks	See official syllabus
CO4	Outline services and protocols of network, transport layers 15 Understanding and application layer	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline basic structure of a computer, motherboard components in desktop	4	As specified
Module 2	Describe power supply and storage devices of a computer	3	As specified
Module 3	Describe the basic concepts of data communication and computer networks	5	As specified
Module 4	layer	4	As specified

MODULE 1

Outline basic structure of a computer, motherboard components in desktop

This module is presented from the official syllabus scope for Networks. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline basic structure of a computer, motherboard components in desktop
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ Illustrate the functional units of a computer with block diagram.

Illustrate the functional units of a computer with block diagram. is an official topic in Module 1 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the functional units of a computer with block diagram. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the functional units of a computer with block diagram. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Illustrate the functional units of a computer with block diagram. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Illustrate the functional units of a computer with block diagram. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Describe different connectors available in desktop computers

Describe different connectors available in desktop computers is an official topic in Module 1 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe different connectors available in desktop computers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe different connectors available in desktop computers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Describe different connectors available in desktop computers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe different connectors available in desktop computers □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain various motherboard components and its features in desktop computer

Explain various motherboard components and its features in desktop computer is an official topic in Module 1 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various motherboard components and its features in desktop computer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various motherboard components and its features in desktop computer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Explain various motherboard components and its features in desktop computer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain various motherboard components and its features in desktop computer [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ **Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS: Define booting & understand the concept of multi os booting, BIOS – settings & configuration**

Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS: Define booting & understand the concept of multi os booting, BIOS – settings & configuration is an official topic in Module 1 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS:

Define booting & understand the concept of multi os booting, BIOS – settings & configuration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe bios settings 4 understanding functional units of a computer- block diagram. desktop motherboard components: motherboard form factors, chipset, cpu socket types, cpu fan & heatsink mounting points, connectors for integrated peripherals, cmos backup battery, bus & interconnection slots, south bridge, north bridge, agp slots, ide & sata connectors- features, memory slots, switches & jumpers. bios: define booting & understand the concept of multi os booting, bios – settings & configuration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS: Define booting & understand the concept of multi os booting, BIOS – settings & configuration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS: Define booting & understand the concept of multi os booting, BIOS – settings & configuration □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Describe power supply and storage devices of a computer

This module is presented from the official syllabus scope for Networks. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe power supply and storage devices of a computer
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▾ Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02

Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02 is an official topic in Module 2 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe characteristics of power supplies. 3 remembering explain the details about powering the pc 5 understanding m2.02 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ List memory hierarchy 2 Understanding Discuss different types of primary memory and

List memory hierarchy 2 Understanding Discuss different types of primary memory and is an official topic in Module 2 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List memory hierarchy 2 Understanding Discuss different types of primary memory and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list memory hierarchy 2 understanding discuss different types of primary memory and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What List memory hierarchy 2 Understanding Discuss different types of primary memory and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ List memory hierarchy 2 Understanding Discuss different types of primary memory and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices.

4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices. is an official topic in Module 2 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct

access, Random access, Sequential access storage devices. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding secondary memory devices 1 purpose and characteristics of power supplies- power connectors- replacement of power supplies, overview of smps and its features. ups: block diagram, types of ups. introduction to computer memory system, memory characteristics & hierarchy. different types of memory – primary memory: ram, rom, cache memory concepts & levels-secondary memory – direct access, random access, sequential access storage devices. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels-Secondary memory – Direct access, Random access, Sequential access storage devices. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe the basic concepts of data communication and computer networks

This module is presented from the official syllabus scope for Networks. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe the basic concepts of data communication and computer networks
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding communication

1 Understanding communication is an official topic in Module 3 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding communication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 1 Understanding communication [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▾ Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4

Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4 is an official topic in Module 3 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare types of networks and topology 2 understanding explain the layered concept of tcp/ip and osi 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Understanding models

Understanding models is an official topic in Module 3 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding models using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding models should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding models means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Understanding models □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and

Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and is an official topic in Module 3 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the concept of physical layer 4 understanding outline the basic concepts of data link layer and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 4 Understanding services Introduction to data communication – definition, components, data representations, data flow Networks – definition, network criteria, types of connection, physical topology, network types, Internet. TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\ Physical layer – Transmission modes, Transmission media – Guided media, Unguided media. Datalink layer –Datalink control- framing, list the types of framing, flow and error control. Datalink layer protocols – For noiseless channel- simplest, stop -and- wait. Outline services and protocols of network, transport layers and application

4 Understanding services Introduction to data communication – definition, components, data representations, data flow Networks – definition, network criteria, types of connection, physical topology, network types, Internet. TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\ Physical layer –Transmission modes, Transmission media – Guided media, Unguided media. Datalink layer –Datalink control- framing, list the types of framing, flow and error control. Datalink layer protocols – For noiseless channel- simplest, stop -and- wait. Outline services and protocols of network, transport layers and application is an official topic in Module 3 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding services Introduction to data communication – definition, components, data representations, data flow

Networks – definition, network criteria, types of connection, physical topology, network types, Internet. TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\ Physical layer – Transmission modes, Transmission media – Guided media, Unguided media. Datalink layer –Datalink control- framing, list the types of framing, flow and error control. Datalink layer protocols – For noiseless channel- simplest, stop -and- wait. Outline services and protocols of network, transport layers and application using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding services introduction to data communication – definition, components, data representations, data flow networks – definition, network criteria, types of connection, physical topology, network types, internet. tcp/ip and osi models- functions of layers in both models, comparison, tcp/ip protocol suite.\ physical layer –transmission modes, transmission media – guided media, unguided media. datalink layer –datalink control- framing, list the types of framing, flow and error control. datalink layer protocols – for noiseless channel- simplest, stop -and- wait. outline services and protocols of network, transport layers and application should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding services Introduction to data communication – definition, components, data representations, data flow Networks – definition, network criteria, types of connection, physical topology, network types, Internet. TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\ Physical layer – Transmission modes, Transmission media – Guided media, Unguided media. Datalink layer –Datalink control- framing, list the types of

Aspect	What to prepare
	framing, flow and error control. Datalink layer protocols – For noiseless channel- simplest, stop -and- wait. Outline services and protocols of network, transport layers and application means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding services Introduction to data communication – definition, components, data representations, data flow Networks – definition, network criteria, types of connection, physical topology, network types, Internet. TCP/IP and OSI models- functions of layers in both models, comparison, TCP/IP protocol suite.\ Physical layer – Transmission modes, Transmission media – Guided media, Unguided media. Datalink layer –Datalink control- framing, list the types of framing, flow and error control. Datalink layer protocols – For noiseless channel- simplest, stop -and- wait. Outline services and protocols of network, transport layers and application □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

layer

This module is presented from the official syllabus scope for Networks. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: layer
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding Control protocols and functions 4

4 Understanding Control protocols and functions 4 is an official topic in Module 4 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Control protocols and functions 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding control protocols and functions 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Control protocols and functions 4 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding Control protocols and functions 4 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▾ Summarize transport layer services and protocols. Understanding Describe the features and services of application 4

Summarize transport layer services and protocols. Understanding Describe the features and services of application 4 is an official topic in Module 4 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize transport layer services and protocols. Understanding Describe the features and services of application 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize transport layer services and protocols. understanding describe the features and services of application 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Summarize transport layer services and protocols. Understanding Describe the features and services of application 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize transport layer services and protocols. Understanding Describe the features and services of application 4 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ Understanding layer. 3

Understanding layer. 3 is an official topic in Module 4 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding layer. 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding layer. 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding layer. 3 means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Understanding layer. 3 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format. Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP & HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777

Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format. Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment

format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP & HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777 is an official topic in Module 4 of Networks. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format. Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP & HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain application layer protocols understanding 1 network layer- network layer services. network layer protocols – internet protocol, ip v4 protocol & ipv6 protocol- packet format. transport layer- transport layer services transport layer protocols- udp-user datagram, features & services, applications. tcp – features & services, applications, segment format. application layer - application layer services. application layer paradigms- client-server, peer to peer application layer protocols – ftp & http - basic models. text/reference t/r book title/author stephen j.bigelow, troubleshooting, maintaining and repairing pcs, tmh, t1 fifth edition. t2 morris rosenthal, an introduction to troubleshooting and repairing laptop. t3 d balasubramanian., computer installation and servicing. tmh, 2010 behrouz a. forouzan -data communications and networking – mcgraw hill edn. – r1 fourth edition/fifth edition complete a+ guide to it hardware and software: aa comptia a+ core 1 (220- r2 1001) & comptia a+ core 2 (220-1002) r3 joel rosenthal , pc repair and maintenance- fire wall media, first edition manahar lotia, pradeep niar, modern computer hardware course bpb r4 publication ,2011 andrew s. tanenbaum ,david j. wetherall computer networks – prentice hall fifth r5 edition r6 william stalling - data communication & networks - prentice hall-tenth edition fred halsall -data communications, computer networks and open systems - r7 addison-wesley, 1996 r5 principles of electronic communications –pradeep kumar ghosh online resources: sl.no. website link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format. Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP

Aspect	What to prepare
	& HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format.

Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP & HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Illustrate the functional units of a computer with block diagram.. (3 marks)**

▶ **Define or explain Describe different connectors available in desktop computers. (3 marks)**

▶ **Define or explain Explain various motherboard components and its features in desktop computer. (3 marks)**

▶ **Define or explain Describe BIOS settings 4 Understanding Functional units of a computer- Block diagram. Desktop Motherboard components: Motherboard form factors, Chipset, CPU Socket types, CPU fan & heatsink mounting points, Connectors for integrated peripherals, CMOS Backup battery, Bus & Interconnection slots, South bridge, North bridge, AGP slots, IDE & SATA Connectors- features, Memory slots, Switches & Jumpers. BIOS: Define booting & understand the concept of multi os booting, BIOS – settings & configuration. (3 marks)**

Module 2

▶ **Define or explain Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02. (3 marks)**

▶ **Define or explain List memory hierarchy 2 Understanding Discuss different types of primary memory and. (3 marks)**

▶ **Define or explain 4 Understanding secondary memory devices 1 Purpose and characteristics of Power Supplies- Power connectors- Replacement of Power Supplies, Overview of SMPS and its features. UPS: Block diagram, Types of UPS. Introduction to computer memory system, Memory characteristics & Hierarchy. Different types of memory – Primary memory: RAM, ROM, Cache memory concepts & levels- Secondary memory – Direct access, Random access, Sequential access storage devices.. (3 marks)**

Module 3

▶ **Define or explain 1 Understanding communication. (3 marks)**

▶ **Define or explain Compare types of networks and topology 2 Understanding Explain the layered concept of TCP/IP and OSI 4. (3 marks)**

▶ **Define or explain Understanding models. (3 marks)**

▶ **Define or explain Describe the concept of physical layer 4 Understanding Outline the basic concepts of data link layer and. (3 marks)**

Module 4

▶ **Define or explain 4 Understanding Control protocols and functions 4. (3 marks)**

▶ **Define or explain Summarize transport layer services and protocols. Understanding Describe the features and services of application 4. (3**

marks)

▶ **Define or explain Understanding layer. 3. (3 marks)**

▶ **Define or explain Explain Application layer protocols Understanding 1 Network layer- Network layer services. Network layer protocols – Internet protocol, IP v4 protocol & IPv6 protocol- Packet Format. Transport layer- Transport layer services Transport layer protocols- UDP-User Datagram, Features & Services, Applications. TCP – Features & Services, Applications, Segment format. Application layer - Application layer services. Application layer Paradigms- client-server, peer to peer Application layer protocols – FTP & HTTP - Basic models. Text/Reference T/R Book Title/Author Stephen J.Bigelow, Troubleshooting, Maintaining and Repairing PCs, TMH, T1 Fifth Edition. T2 Morris Rosenthal, An introduction to Troubleshooting and Repairing laptop. T3 D Balasubramanian., Computer installation and servicing. TMH, 2010 Behrouz A. Forouzan -Data Communications and Networking – McGraw Hill Edn. – R1 Fourth Edition/Fifth Edition Complete A+ Guide to IT Hardware and Software: AA CompTIA A+ Core 1 (220- R2 1001) & CompTIA A+ Core 2 (220-1002) R3 Joel Rosenthal , PC Repair and Maintenance- Fire wall Media, First Edition Manahar Lotia, Pradeep Niar, Modern Computer Hardware Course BPB R4 Publication ,2011 Andrew S. Tanenbaum ,David J. Wetherall Computer Networks – Prentice Hall fifth R5 Edition R6 William Stalling - Data Communication & Networks - Prentice Hall-Tenth Edition Fred Halsall -Data Communications, Computer Networks and Open Systems - R7 Addison-Wesley, 1996 R5 Principles of Electronic Communications –Pradeep Kumar Ghosh Online resources: Sl.No. Website Link 1 https://nptel.ac.in/courses/117101051/ 2 https://www.tutorialspoint.com/digital_communication 3 https://nptel.ac.in/courses/108101113/ 4 https://nptel.ac.in/courses/117/105/117105143/ 5 https://www.slideshare.net/bhavyaw/noise-13468777. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Illustrate the functional units of a computer with block diagram.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe characteristics of Power Supplies. 3 Remembering Explain the details about powering the PC 5 Understanding M2.02**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **1 Understanding communication**.

Module 4

1-mark definition: State the meaning of **4 Understanding Control protocols and functions 4**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link https://nptel.ac.in/courses/117101051/
- https://www.tutorialspoint.com/digital_communication https://nptel.ac.in/courses/108101113/
- https://nptel.ac.in/courses/117/105/117105143/
- https://www.slideshare.net/bhavyaw/noise-13468777

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 6371A. Verify institutional updates against the current official curriculum.